SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – CLASS TEST (5 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2– Manipulation (BL3)	Assessment:	Assessment Tool 3 – Class Test
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	*Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)*		
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Section / Batch:	SOITT - D	Date:	01 - 08 - 26

Code of Conduct (192472332)

I Chavali Chandu *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Chuy.

Instructions

- This test contains 5 questions, each carrying 10 marks. Total: 50 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Image Enhancement in Spatial Domain	Improving image quality using spatial domain techniques	1	10
Histogram Processing Techniques	Enhancing image contrast through histogram equalization, histogram specification, and histogram stretching.	2	10
Spatial Filtering	Noise reduction and feature enhancement	3	10
Image Enhancement in Frequency Domain	Enhancing images using frequency domain transformations	4	10
Image Segmentation	Partitioning images into meaningful regions	5	10
		GRAND TOTAL:	**50 Marks**

$CO_2 - AT_3$
Class TEST.

Image Enhancement in the Spatial Domain

20/25

Image Enhancement in the Spatial Domain

Definition:

→ Image enhancement is the process of improving the visual quality of an image so that important details become clearer & easier to analyze.

Basic Gray-level Transformations:

Gray-level transformations change the intensity values of pixel.

Common transformation include:

→ Image Negative: Converts dark areas into dark areas.

→ log Transformations: Expands pixel values & compresses bright values.

→ Power law Transformation: Adjusts image brightness & contrast using gamma value.

→ Contrast stretching: Expands a narrow range of intensity values to improve contrast.

Histogram processing:

A histogram represents the distribution of pixel intensity values to improve contrast.

Histogram Equalization: Redistributes intensity values to improve contrast.

→ Histogram stretching: Expands the intensity range of an image, making dark & bright regions more distinguishable.

→ These techniques are suitable when the captured image has poor contrast or uneven intensity distribution.

Spatial Filtering:

Spatial filtering processes pixel using a small neighborhood called a filter mask or kernel.

→ Smoothing filters: Reduce noise & unwanted variations.

→ Sharpening filters: Enhance edges & fine details.

Technique	Suitable for
Gray level transformation	Brightness & intensity correction

Histogram processing	Improving contrast
Smoothing filter	Removing noise
Sharphening filter	Enhancing edges & details.

Example: A low-quality medical image may contain noise and poor contrast Histogram can improve contrast, smoothing & reduce noise, & sharphening can make important boundaries clearer.

Smoothing & Sharpending:

- Smoothing removes unwanted noise from an image.
  - → Makes image smooth.
  - → Used for noise removal.
  - → Example: Mean filter, Median filter.
- Sharphening makes edges & details clearer.
  - → Makes image more clear.
  - → Used for edge enhancement.
  - → Example: laplacian filter.

Difference:

→ Smoothing → Remove noise.

→ sharphening → Improves edges.

Conclusion: Smoothing is used for noise removal, while Sharphening is used for detail enhancement.

3. Frequency Domain Enhancement:

→ It improves an image using frequency Components.

The image is changed using the Fourier transform.

low-pass filter:

→ Remove noise

→ Makes image smooth.

→ May blur edges.

High-pass filter:

→ Makes edges clear.

→ Improve details.

→ Makes image sharp.

difference:

low pass → smoothing

high pass → Sharphening.

Conclusion: Frequency filters are used to improve the image quality.

Image Segmentation:

It means dividing an image into different meaningful parts.

Types:

1. point detection :- find individual points.
2. Line detection - find lines.
3. Edge detection - find object edges.
4. Thresholding - seperates object from background.
5. Region - based segmentation: - groups similar pixels.

Example: separating a person from the background in a photo.

Conclusion: Segmentation helps to identify objects easily.

5. Edge Detection, Edge Linking & Boundary Detection.

Edge detection : Finds the edges of objects in an image.

Edge linking : Connects broken edges together.

Boundary detection: Finds the complete boundary of an object.

Example: In a damaged metal part.

→ Edge detection finds the crack.

→ Edge linking connects the crack

→ Bounday detection find the complete crack shape.

Uses:

→ Object recognition.

→ Shape detection.

→ Defect detection.

Conclusion: These techniques help computers find an recognize objects and defects.